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  <title>MTH102 Complete Notes (All 9 Modules)</title>
  <style>
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      font-weight: 600;
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```



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}
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    .module-box { break-inside: avoid; }
}
</style>
</head>
<body>

<h1>MTH102: Elementary Mathematics II</h1>
<div class="subtitle">Complete Lecture Notes & Q&A — All 9 Modules</div>
<p style="text-align:center; color:#4a5c5c;">Compiled by Sunday Ojodomo Michael |
Federal University Lokoja</p>
<hr>

<div class="nav-syllabus">
    <h3>    Course Syllabus (Modules 1–9)</h3>
    <ul>
        <li><strong>Module 1:</strong> Functions of Real Variables (Domain, Range,

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Even/Odd, Periodic, Composition, Difference Quotient)

Module 2: Limits & Limit Theorems (Intuition, Theorems, Squeeze Theorem, Infinite Limits, Indeterminate Forms)

Module 3: Continuity (Definition, Types of Discontinuities, IVT, Inverse Functions)

Module 4: Differentiation from First Principles (Definition, Power/Sum Rules, Derivative of $\sin x$)

Module 5: Advanced Differentiation (Product, Quotient, Chain Rule, Trig/Log/Exp, Implicit, Higher-order)

Module 6: Applications of Derivatives (Increasing/Decreasing, Max/Min, Optimization)

Module 7: Integration as Anti-differentiation (Indefinite Integrals, Power Rule, u-Substitution)

Module 8: Advanced Integration Techniques (Parts, Partial Fractions, Trig Substitution)

Module 9: The Fundamental Theorem of Calculus (Definite Integrals, Area under & between curves)

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<!-- MODULE 1 -->

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<h2> Module 1: Functions of Real Variables</h2>

<div class="module-box">

<h4>Core Concepts</h4>

<p>Definition: A function f from set X (Domain) to set



$f: X \rightarrow Y$ (Codomain) assigns **exactly one** $f(x) \in Y$ to every $x \in X$.

Domain:

All valid inputs.

Range: Actual outputs. ($\text{Range} \subseteq \text{Codomain}$).

Even: $f(-x) = f(x)$ (Symmetry over Y-axis).

Odd: $f(-x) = -f(x)$ (Rotational symmetry).

Periodic: $f(x+T) = f(x)$. Fundamental period T .

Composite: $(f \circ g)(x) = f(g(x))$. Domain: $x \in D_g$ and $g(x) \in D_f$.

Difference Quotient: $\frac{f(x+h)-f(x)}{h}$ (Slope of secant line).

Q1 (Domain & Range): Find the domain and range of $f(x) = \frac{3x}{x^2 - 4}$.

Q2 (Even/Odd): Determine if $f(x) = x^3 \sin(x)$ is even, odd, or neither.

Q3 (Composite): Given $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x-2}$, find $(f \circ g)(x)$ and its domain.

Q4 (Periodicity): What is the fundamental period of $f(x) =$

$\cos(4x)$?

Q5 (Difference Quotient): Compute $\frac{f(x+h)-f(x)}{h}$ for $f(x) = 2x^2 - 3x$. Simplify.

Solutions:

A1: Domain: $x \neq \pm 2$. Range: $\mathbb{R}$ (because discriminant $9+16y^2 \geq 0$).

A2: $f(-x) = (-x)^3 \sin(-x) = (-x^3)(-\sin x) = x^3 \sin x = f(x) \rightarrow$
Even.

A3: $(f \circ g)(x) = \frac{1}{\sqrt{x-2}}$. Domain: $x > 2$.

A4: $T = \frac{2\pi}{4} = \frac{\pi}{2}$.

A5: $f(x+h) = 2x^2+4xh+2h^2-3x-3h$. Subtract $f(x)$:
 $4xh+2h^2-3h$. Divide by h : $4x + 2h - 3$.

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MODULE 2 -->

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Module 2: Limits & Limit Theorems

Core Concepts

Intuition: A limit $\lim_{x \rightarrow a} f(x) = L$ means as x gets arbitrarily close to a (from both sides), $f(x)$ gets arbitrarily close to L .

Limit Theorems: Constant, Sum, Product, Quotient (provided denominator $\neq 0$), Power, Root.

Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ and $\lim g = \lim h = L$, then $\lim f = L$.

Infinite Limits: $\lim_{x \rightarrow a} \frac{1}{(x-a)^2} = +\infty$.

Indeterminate Forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$. Solve via factorization or rationalization.

Q1: Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

Q2: Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 4}$.

Q3: Use the Squeeze Theorem: If $1 - \frac{x^2}{2} \leq f(x) \leq 1 + \frac{x^2}{2}$, find $\lim_{x \rightarrow 0} f(x)$.

Q4: Evaluate $\lim_{x \rightarrow 0} \frac{\sin(2x)}{x}$.

<div class="answer-box">

Solutions:

A1: Factor: $\frac{(x-3)(x+3)}{x-3} = x+3$. Limit = $\lim_{x \rightarrow 3} (x+3) = \mathbf{6}$.

A2: Divide numerator and denominator by x^2 : $\frac{3 + \frac{2}{x}}{5 - \frac{4}{x^2}} \rightarrow \frac{3}{5}$.

A3: $\lim_{x \rightarrow 1} (1 - x^2/2) = 1$, $\lim_{x \rightarrow 1} (1 + x^2/2) = 1$. Therefore, $\lim_{x \rightarrow 1} f(x) = \mathbf{1}$.

A4: Use the standard limit $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$. Let $u = 2x$: $\lim_{x \rightarrow 0} \frac{\sin(2x)}{2x} \cdot 2 = 1 \cdot 2 = \mathbf{2}$.

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<!-- MODULE 3 -->

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<h2> Module 3: Continuity</h2>

<div class="module-box">

<h4>Core Concepts</h4>

<p>A function is continuous at $x = a$ if three conditions are met:</p>

 $f(a)$ is defined.

 $\lim_{x \rightarrow a} f(x)$ exists.

 $\lim_{x \rightarrow a} f(x) = f(a)$.



Types of Discontinuities: Removable (hole), Jump (step), Infinite (vertical asymptote).

Intermediate Value Theorem (IVT): If f is continuous on $[a, b]$ and k is between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = k$.

Q1: Is $f(x) = \frac{x^2-4}{x-2}$ continuous at $x=2$? If not, how can it be redefined to make it continuous?

Q2: Find the value of k that makes $f(x) = \begin{cases} x^2 + k, & x \leq 1 \\ 2x + 3, & x > 1 \end{cases}$ continuous at $x=1$.

Q3: Show that $f(x) = x^3 - x - 1$ has a root between 1 and 2 using IVT.

Solutions:

A1: No. $f(2)$ is undefined. Factor: $\frac{(x-2)(x+2)}{x-2} = x+2$. Redefine $f(2) = 4$ to make it continuous.

A2: Left limit: $1^2 + k = 1+k$. Right limit: $2(1)+3 = 5$. Set $1+k = 5 \Rightarrow \mathbf{k=4}$.

A3: $f(1) = 1-1-1 = -1$ (negative). $f(2) = 8-2-1 = 5$ (positive). Since 0 is between -1 and 5 , by IVT, there is a root $c \in (1, 2)$.

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<!-- MODULE 4 -->

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<h2> Module 4: Differentiation from First Principles</h2>

<div class="module-box">

<h4>Core Concepts</h4>

<p>Definition of Derivative: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.</p>

<p>Rules derived from First Principles:</p>

Constant rule: $\frac{d}{dx}(c) = 0$.

Power rule: $\frac{d}{dx}(x^n) = nx^{n-1}$.

Sum rule: $\frac{d}{dx}(f+g) = f' + g'$.

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<div class="test-question">

Q1: Using first principles, find the derivative of $f(x) = x^2$.

</div>

<div class="test-question">

Q2: Using first principles, find the derivative of $f(x) = \sqrt{x}$.

</div>

<div class="test-question">

Q3: Differentiate $f(x) = 5x^4 - 3x^2 + 7$ using the power rule.

</div>

<div class="answer-box">

Solutions:

A1: $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = \mathbf{2x}$.

A2: $f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$.
Rationalize: $\frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \rightarrow \frac{1}{2\sqrt{x}}$.

A3: $f'(x) = 20x^3 - 6x$.

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<!-- MODULE 5 -->

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<h2> Module 5: Advanced Differentiation</h2>

<div class="module-box">

<h4>Core Concepts</h4>

Product Rule: $(uv)' = u'v + uv'$.

Quotient Rule: $(u/v)' = \frac{u'v - uv'}{v^2}$.

Chain Rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.

Derivatives of Special Functions: $\frac{d}{dx} \sin x = \cos x$, $\frac{d}{dx} \cos x = -\sin x$, $\frac{d}{dx} e^x = e^x$, $\frac{d}{dx} \ln x = \frac{1}{x}$.



Implicit Differentiation: Differentiate both sides with respect to x and solve for $\frac{dy}{dx}$.

Q1:

Differentiate $y = (3x^2 + 1)^5$ (Chain Rule).

Q2:

Differentiate $y = x^2 \sin(x)$ (Product Rule).

Q3:

Differentiate $y = \frac{\ln x}{x^2}$ (Quotient Rule).

Q4:

Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$ (Implicit Differentiation).

Solutions:

A1: Let $u = 3x^2 + 1$. $\frac{dy}{du} = 5u^4$, $\frac{du}{dx} = 6x$. $\frac{dy}{dx} = 5(3x^2 + 1)^4 \cdot 6x = \mathbf{30x(3x^2 + 1)^4}$.

A2: $u = x^2, u' = 2x$; $v = \sin x, v' = \cos x$. $y' = 2x \sin x + x^2 \cos x$.

A3: $u = \ln x, u' = 1/x$; $v = x^2, v' = 2x$. $y' = \frac{(1/x)x^2 - (\ln x)2x}{(x^2)^2} = \frac{x - 2x \ln x}{x^4} = \mathbf{\frac{1 - 2 \ln x}{x^3}}$.

A4: Differentiate: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow$

$\frac{dy}{dx} = \mathbf{-\frac{x}{y}}$.

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<!-- MODULE 6 -->

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<h2> Module 6: Applications of Derivatives</h2>

<div class="module-box">

<h4>Core Concepts</h4>

Increasing/Decreasing: If $f'(x) > 0$, f is increasing. If $f'(x) < 0$, f is decreasing.

Critical Points: Where $f'(x) = 0$ or undefined.

First Derivative Test: Sign change of f' around critical points determines local max/min.

Second Derivative Test: If $f''(x) > 0$, local min; if $f''(x) < 0$, local max.

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<div class="test-question">

Q1: Find the critical points of $f(x) = x^3 - 3x^2 - 9x + 5$ and classify them.

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<div class="test-question">

Q2: Find the intervals where $f(x) = x^3 - 3x$ is increasing and

decreasing.

</div>

<div class="test-question">

Q3 (Optimization): Find two positive numbers whose sum is 20 and whose product is maximum.

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<div class="answer-box">

Solutions:

A1: $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$. Critical points: $(x = -1, 3)$.
 $f''(x) = 6x - 6$. At $(x = -1)$, $f'' = -12 < 0 \rightarrow$ Local Max. At $(x = 3)$, $f'' = 12 > 0 \rightarrow$ Local Min.

A2: $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. Increasing on $(-\infty, -1)$ and $(1, \infty)$. Decreasing on $(-1, 1)$.

A3: Let numbers be (x) and $(20-x)$. Product $(P = x(20-x) = 20x - x^2)$. $P' = 20 - 2x = 0 \rightarrow x = 10$. $P'' = -2 < 0 \rightarrow$ Max. Numbers are 10 and 10.

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<!-- MODULE 7 -->

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<h2> Module 7: Integration as Anti-differentiation</h2>

<div class="module-box">

<h4>Core Concepts</h4>

<p>Indefinite Integral: $\int f(x) dx = F(x) + C$, where $(F'(x) = f(x))$.</p>



Power Rule (Reverse): $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, for $(n \neq -1)$.

Special Cases: $\int \frac{1}{x} dx = \ln|x| + C$, $\int e^x dx = e^x + C$.

u-Substitution: Reverse of the Chain Rule. Set $(u = g(x))$, $(du = g'(x) dx)$.

</div>

<div class="test-question">

Q1: Evaluate $\int (3x^2 + 4x - 1) dx$.

</div>

<div class="test-question">

Q2: Evaluate $\int \frac{2x}{x^2 + 1} dx$ (Use u-substitution).

</div>

<div class="test-question">

Q3: Evaluate $\int e^{3x} dx$.

</div>

<div class="answer-box">

Solutions:

A1: $3 \cdot \frac{x^3}{3} + 4 \cdot \frac{x^2}{2} - x + C = \mathbf{x^3 + 2x^2 - x + C}$.

A2: Let $(u = x^2 + 1)$, $(du = 2x dx)$. Integral becomes $\int \frac{1}{u} du = \ln|u| + C = \mathbf{\ln(x^2 + 1) + C}$.

A3: Let $(u = 3x)$, $(du = 3 dx \Rightarrow dx = du/3)$. $\int e^u \frac{du}{3} = \frac{1}{3}e^u + C = \mathbf{\frac{1}{3}e^{3x} + C}$.

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<!-- MODULE 8 -->

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<h2> Module 8: Advanced Integration Techniques</h2>

<div class="module-box">

<h4>Core Concepts</h4>

Integration by Parts: $\int u \, dv = uv - \int v \, du$. Choose u as the function that simplifies (LIATE rule).

Partial Fractions: Decompose rational functions $\frac{P(x)}{Q(x)}$ into simpler fractions for integration.

Trigonometric Substitution: Use identities like $1 - \sin^2 = \cos^2$ to handle radicals $\sqrt{a^2 - x^2}$.

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<div class="test-question">

Q1: Evaluate $\int x e^x \, dx$ (Integration by Parts).

</div>

<div class="test-question">

Q2: Evaluate $\int \frac{1}{x^2 - 1} \, dx$ (Partial Fractions).

</div>

<div class="test-question">

Q3: Evaluate $\int x \cos(x) \, dx$ (Integration by Parts).

</div>

<div class="answer-box">

Solutions:

A1: Let $(u = x, dv = e^x dx \rightarrow du = dx, v = e^x)$. $\int x e^x dx = x e^x - \int e^x dx = \mathbf{x e^x - e^x + C}$.

A2: Decompose: $\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$. Integral = $\frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C = \mathbf{\frac{1}{2} \ln|\frac{x-1}{x+1}| + C}$.

A3: Let $(u = x, dv = \cos x dx \rightarrow du = dx, v = \sin x)$. $\int x \cos x dx = x \sin x - \int \sin x dx = \mathbf{x \sin x + \cos x + C}$.

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<!-- MODULE 9 -->

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<h2> Module 9: The Fundamental Theorem of Calculus & Area</h2>

<div class="module-box">

<h4>Core Concepts</h4>

FTC Part 1: $\frac{d}{dx} \int_a^x f(t) dt = f(x)$.
(Differentiation undoes integration).

FTC Part 2: $\int_a^b f(x) dx = F(b) - F(a)$, where F is the antiderivative.

Area under curve: $\int_a^b f(x) dx$ (if $f \geq 0$ on $[a,b]$).

Area between curves: $\int_a^b (\text{Top} - \text{Bottom}) dx$.

</div>

<div class="test-question">

Q1: Evaluate $\int_0^2 (x^3 - x) dx$.

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<div class="test-question">

Q2: Find the area under $f(x) = x^2$ from $x=1$ to $x=3$.

</div>

<div class="test-question">

Q3: Find the area between the curves $y = x$ and $y = x^2$.

</div>

<div class="test-question">

Q4: If $F(x) = \int_0^x \sqrt{t^2 + 1} dt$, find $F'(2)$.

</div>

<div class="answer-box">

Solutions:

A1: Antiderivative: $\frac{x^4}{4} - \frac{x^2}{2}$. Evaluate from 0 to 2: $(\frac{16}{4} - \frac{4}{2}) - (0) = (4 - 2) = \mathbf{2}$.

A2: $\int_1^3 x^2 dx = \left[\frac{x^3}{3} \right]_1^3 = \frac{27}{3} - \frac{1}{3} = \frac{26}{3}$.

A3: Intersection points: $x = x^2 \Rightarrow x=0, 1$. Top is $y = x$, Bottom is $y = x^2$. Area = $\int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \mathbf{\frac{1}{6}}$.

A4: By FTC Part 1, $F'(x) = \sqrt{x^2 + 1}$. Thus, $F'(2) = \sqrt{4 + 1} = \mathbf{\sqrt{5}}$.

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<h2 style="text-align:center; border-bottom: none;"> End of MTH102 Complete
Study Notes</h2>

<p style="text-align:center; color:#4a5c5c;">Prepared with depth, context, and clarity.
Good luck with your studies, Sunday!</p>

</body>

</html>